1. Derivation of I/V characteristics:

In order to obtain the relationship between the drain current of a MOSFET and its terminal voltages, we make two observations.

First, Consider a semiconductor bar carrying a current I (Fig. a). If the charge density along the direction of current is Q_d Coulombs per meter and the velocity of the charge is v meters per second, then



Fig. a

$$I = Q_d \cdot v \quad \ldots (1)$$

To understand why, we measure the total charge that passes through a cross section of the bar in unit-time. With a velocity v, all of the charge enclosed in v meters of the bar must flow through the cross section in one second (Fig. b).

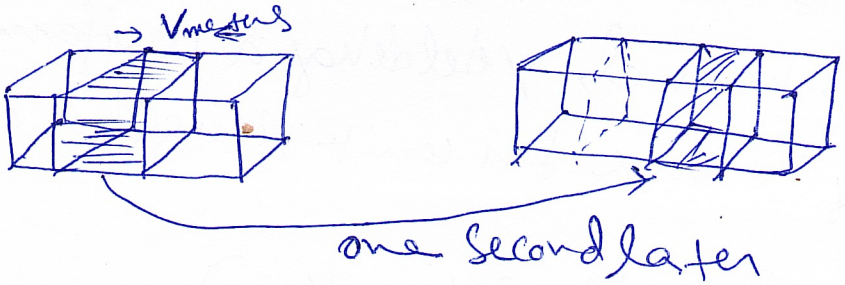
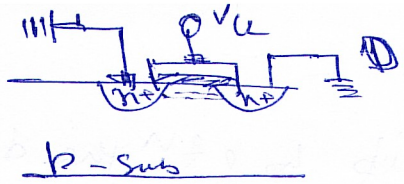
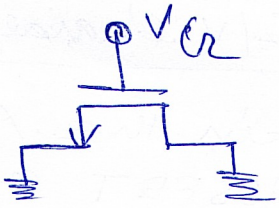


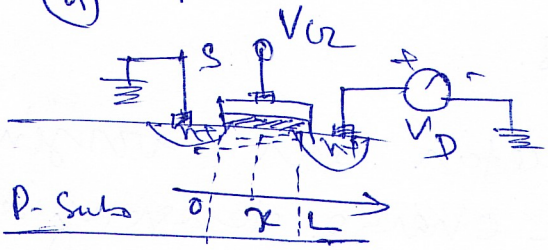
Fig. b

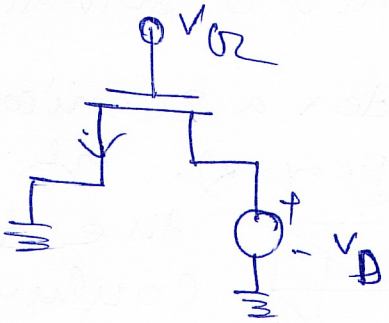
Since the charge density is Q_d, the total charge in v meters equals $Q_d \cdot v$.



(a) Equal source & drain







(b). Unequal source & drain voltages.

Second, consider an NFET whose source and drain are connected to ground (Fig. a). What is the charge density in the inversion layer? Since we assume the onset of inversion occurs at $V_{GS} = V_{TH}$, the inversion charge density produced by the gate oxide capacitance is proportional to $V_{GS} - V_{TH}$. For $V_{GS} \geq V_{TH}$, any charge placed on the gate must be mirrored by the charge in the channel, yielding a uniform channel charge density (charge per unit length) equal to

$$Q_d = WC_{ox}(V_{GS} - V_{TH}) \quad \ldots \quad (2)$$

where C_{ox} is multiplied by W to represent the total capacitance per unit length.

Now suppose, as depicted in Fig (b), the drain voltage is greater than zero. Since the channel potential varies from zero at the source to V_D at the drain, the local voltage difference betn the gate and the channel varies from V_{GS} to $V_{GS} - V_D$.

Thus, the charge density at a point x along the channel can be written as

$$Q_d(x) = W C_{ox}[V_{GS} - V(x) - V_{TH}] \quad \text{---- (3)}$$

where $V(x)$ is the channel potential at x.

Now the current is given by

$$I_D = -W C_{ox}[V_{GS} - V(x) - V_{TH}]\, v \quad \text{---- (4)}$$

where the negative sign is inserted because the charge carriers are negative and v denotes the velocity of the electrons in the channel. For semiconductors, $v = \mu E$, where μ is the mobility of charge carriers and E is the electric field. Noting that $E(x) = -dV/dx$ and μ_n representing the mobility of electrons by μ_n, we have

$$I_D = W C_{ox}[V_{GS} - V(x) - V_{TH}]\, \mu_n \frac{dV(x)}{dx} \quad \text{---- (5)}$$

Subject to boundary condition $V(0) = 0$ and $V(L) = V_{DS}$. Multiplying both sides by dx and performing the integration, we obtain

$$\int_{x=0}^{L} I_D\, dx = \int_{V_{20}}^{V_{DS}} W C_{ox} \mu_n [V_{GS} - V(x) - V_{TH}]\, dV \quad \text{---- (6)}$$

Since I_D is constant along the channel:

$$I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right] \quad \text{---- (7)}$$

Note that L is the effective channel length.

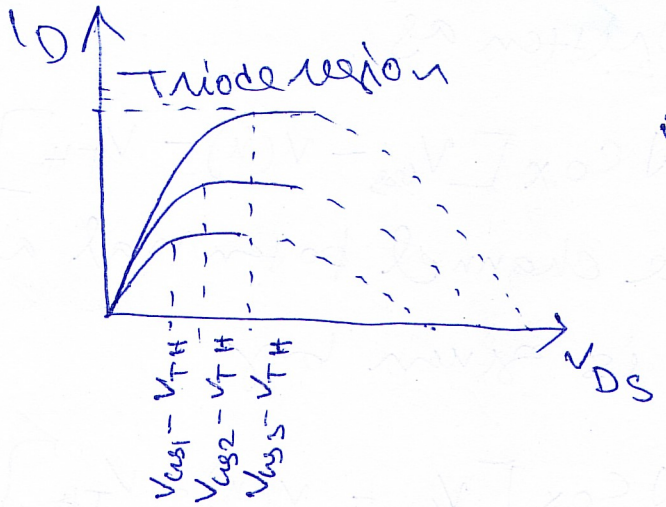


Fig. Drain current versus drain-source voltage in the triode region

Fig. plots the parabolas by eqn (7) for different values of V_{GS}, indicating that the "current capability" of the device increases with V_{GS}. Calculating $\frac{\partial I_D}{\partial V_{DS}}$, the peak of each parabola occurs at $V_{DS} = V_{GS} - V_{TH}$ and the peak current is

$$I_{D,max} = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2 \quad \cdots (8)$$

We call $V_{GS} - V_{TH}$ the "overdrive voltage or sometimes called the "effective voltage" and W/L the aspect ratio." If $V_{DS} \leq V_{GS} - V_{TH}$, we say the device operates in the "triode region".

Note that the integration of eqn (6) assumes μ_n and V_{TH} are independent of x and the gate and drain voltages.

If $V_{DS} \ll 2(V_{GS} - V_{TH})$ we have

$$I_D \approx \mu_n C_{ox}\frac{W}{L}(V_{GS} - V_{TH})V_{DS} \quad \cdots (9)$$

that is, the drain current is a linear function of V_{DS}. For small V_{DS} each parabola can be approximated by a straight line. The linear relationship implies that the path from the source to the drain can be represented by a linear resistor equal to

$$R_{on} = \frac{1}{\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})} \quad \cdots (10)$$

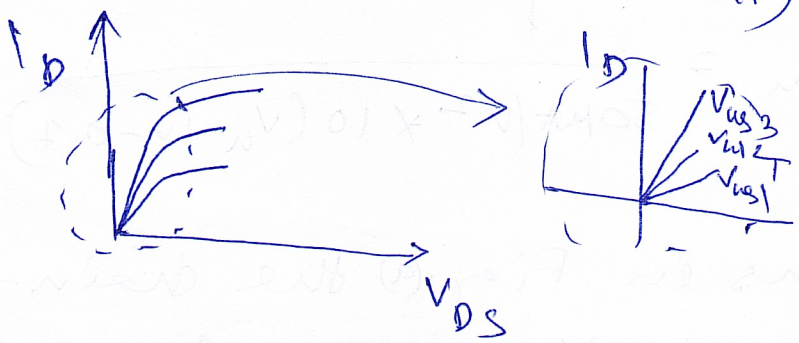


Fig. Linear operation in deep triode region.

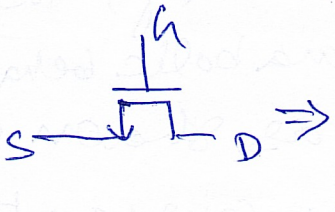
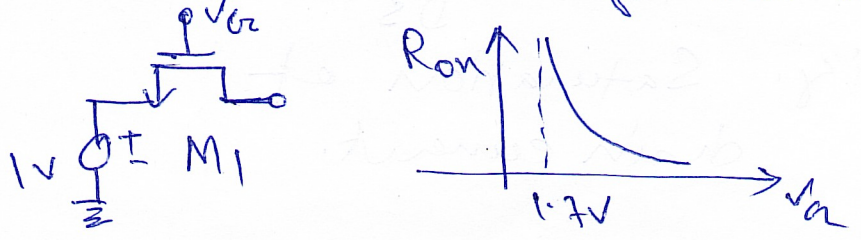


Fig: MOSFET as a controlled linear resistor

A MOSFET can therefore operates as a resistor whose value is controlled by the overdrive voltage. So long as $V_{DS} \ll 2(V_{GS} - V_{TH})$.

Problem: For the arrangement in Fig. below



Plot the on-resistance of M_1 as a function of V_{in}

Assume $\mu_n C_{ox} = 50\,\mu A/V^2$, $W/L = 10$, and $V_{TH} = 0.7V$. Note that the drain terminal is open.

Soln: Since the drain terminal open, $I_D = 0$, and $V_{DS} = 0$. Thus, if the device is on, it operate in the deep triode region. For $V_{GS} < 1V + V_{TH}$, M_1 is off and $R_{on} = \infty$, For ~~$V_{GS} > 1V + V_{TH}$~~, ~~M_1 is off. For~~ $V_{in} > 1V + V_{TH}$, we have

$$R_{on} = \frac{1}{50\mu A/V^2 \times 10 (V_{in} - 1V - 0.7)}$$

What happens in Fig.(a) the drain-source voltage exceeds $V_{GS} - V_{TH}$? In reality, the drain current does not follow the parabolic behaviour for $V_{DS} > V_{GS} - V_{TH}$. In fact, as shown in Fig below, I_D becomes relatively constant and we say the device operates in the saturation region.

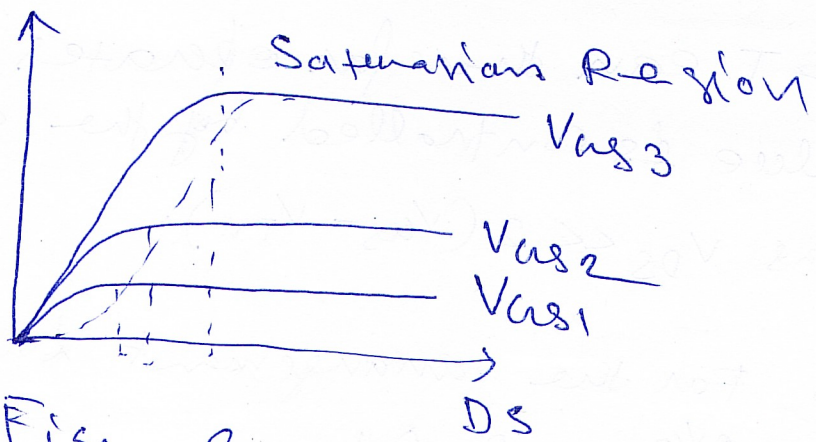


Fig. Saturation of drain current.

From eqn 3. the density of inversion layer charge is proportional to $V_{GS} - V(x) - V_{TH}$. Thus, if $V(x)$ approaches $V_{GS} - V_{TH}$, the $Q_d(x)$ drops to zero. In other words, as depicted in Fig below, if V_{DS} is slightly greater than $V_{GS} - V_{TH}$, then the inversion layer stops at $x \leq L$, and we say the channel is "Pinched off". As V_{DS} increases further, the point at which Q_d equals zero gradually moves toward the source. Thus, at some point along the channel, the local potential difference bet^n the gate and the oxide-silicon interface is not sufficient to support an inversion layer.

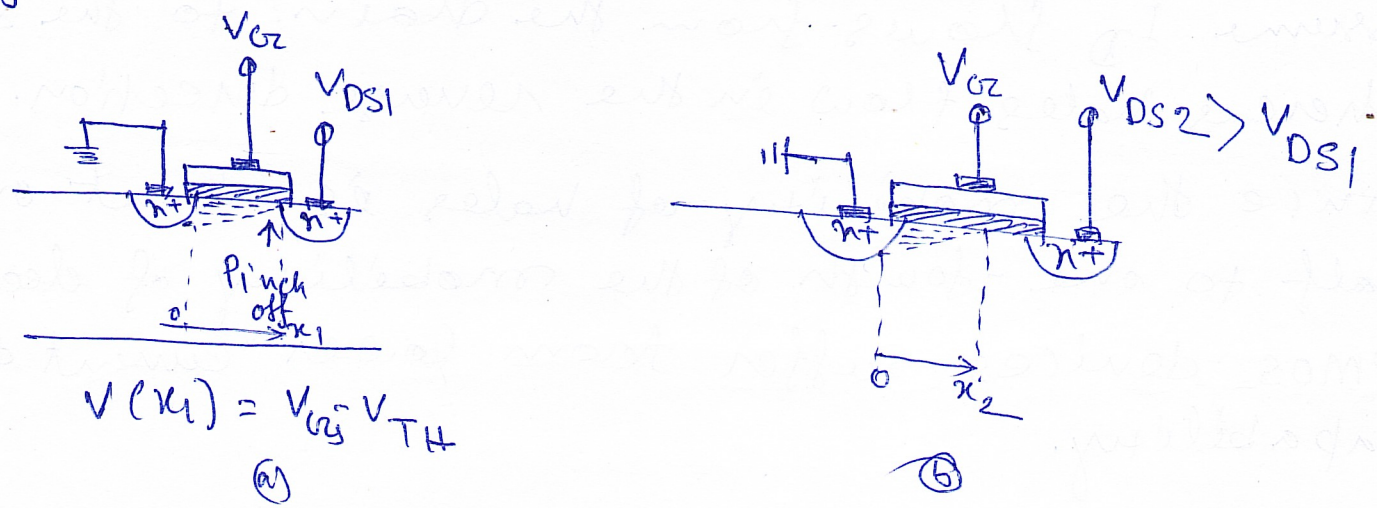


Fig. Pinch-off behavior.

With the above observations, we re-examine eqn (6) for a saturated device. Since Q_d is the density of mobile charge, the integral on the left-hand side of eqn (6) must be taken from $x = 0$ to $x = L'$, where L' is the point at which

Q_2 drops to zero, and that on the right from $V(x) = 0$ to $V(x) = V_{GS} - V_{TH}$. As a result

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L'}(V_{GS} - V_{TH})^2 \quad \text{---(11)}$$

indicating that I_D is relatively independent of V_{DS} if L' remains close to L.

For PMOS devices

$$I_D = -\mu_p C_{ox}\frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \frac{1}{2}V_{DS}^2\right] \quad (12)$$

&

$$I_D = -\frac{1}{2}\mu_p C_{ox}\frac{W}{L'}\left[(V_{GS} - V_{TH})^2\right] \quad (13)$$

The negative sign appears here because we assume I_D flows from the drain to the source, whereas holes flow in the reverse direction. Since the mobility of holes is about one-half to one-fourth of the mobility of electrons, PMOS devices suffer from lower "current drive" capability.

With the approximation $L \approx L'$, a saturated MOSFET can be used as a current source connected bet^n the drain and the source, an important component in analog design. Note that the current sources inject current into ground or draw current from V_{DD}. In other words, only one terminal of each current source is

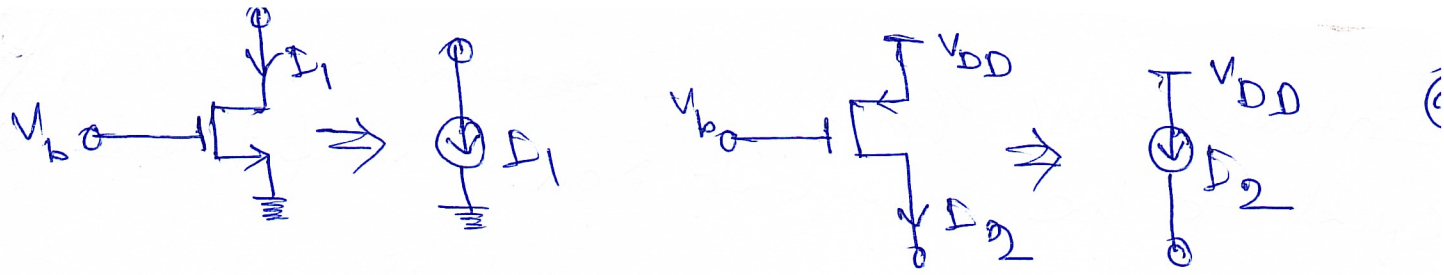


Fig. Saturated MOSFETs operating as current sources.

Since a MOSFET operating in a saturation produces a current in response to its gate-source overdrive voltage, we may define a figure of merit that indicates how well a device converts a voltage and current

More specifically, since in processing signals we deal with the changes in voltages and currents, we define the figure of merit as the change in the drain current divided by the change in the gate-source voltage. Called the "transconductance" and denoted by g_m, this quantity is expressed as

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_{V_{DS}\ \text{const.}} \quad \cdots (14)$$

$$= \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) \quad \cdots (15)$$

In a sense, g_m represents the sensitivity of the device: for a high g_m, a small change in V_{GS} results in a large change in I_D.

Interestingly, g_m in the saturation region is equal to the inverse of R_{on} in deep triode region.

The g_m can be expressed as

$$g_m = \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D} \quad \text{---} \quad (16)$$

$$= \frac{2I_D}{V_{GS} - V_{TH}} \quad \text{---} \quad (17)$$

Plotted in Fig. each of the above expressions proves useful in studying the behaviour of g_m as a function of one parameter while others parameters remain constant.

For example, Fig. (eqn 15), suggests that g_m increases with the overdrive if W/L is constant whereas (17) implies that g_m decreases with overdrive if I_D is constant. The concept of transconductance can also be applied to a device operating in the triode region, as illustrated in the following examples:

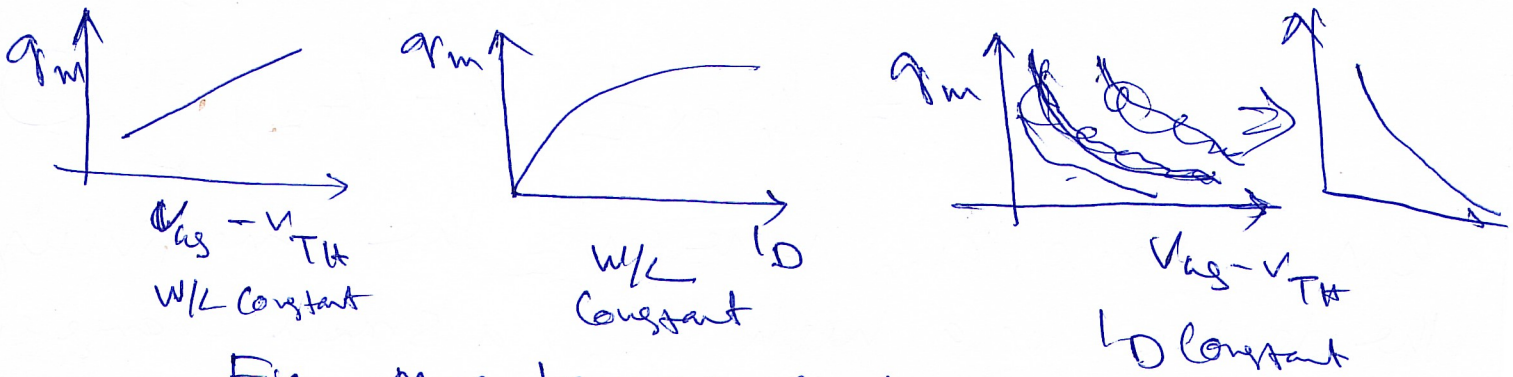


Fig. MOS transconductance as a function of overdrive and drain current.